Comparing 8 RAG Architectures using Python (Anthropic + FAISS)

Retrieval-Augmented Generation (RAG) is one of the most practical ways to give a large language model access to your own data. Here's how 8 popular RAG architectures compare on the same benchmark.

Retrieval-Augmented Generation (RAG) is a popular technique for grounding large language models (LLMs) in your own documents. Instead of relying purely on what the model learned during training, a RAG system retrieves relevant passages at query time and passes them to the LLM as context. However, there are many ways to design a RAG pipeline, and the choice of architecture has a big impact on answer quality, latency, and cost. This tutorial includes the following:

- What is RAG (and why there are so many variants)
- The 8 architectures we will compare
- Setting up the evaluation harness (SQuAD, embeddings, FAISS, metrics)
- Implementing each architecture step-by-step
- Running the benchmark and visualizing the results
- Discussion: when to use which architecture

The code used in this tutorial is available on my GitHub.

What is RAG

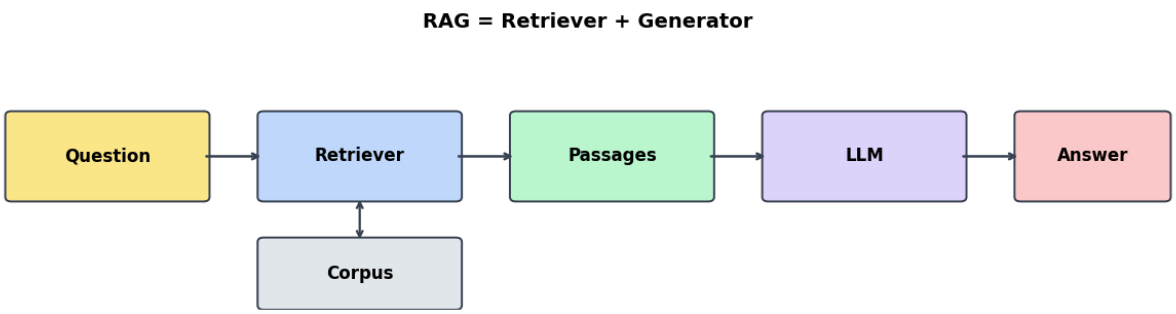
A plain LLM answers questions from the knowledge it picked up during training. That works well for general questions, but it has two obvious problems:

1. The model can't know about anything that happened after its training cutoff.
2. The model doesn't know anything specific to your documents, your company, or your niche.

RAG addresses both problems by splitting the pipeline into two steps. First, a **retriever** looks up the most relevant passages from a corpus you control. Then a **generator** (the LLM) is asked to answer the question using those passages as context. The retriever is typically a vector search index built on top of text embeddings. The diagram below shows the shape of the pipeline:

In [1]:
```
# A small matplotlib schematic of the RAG pipeline. This runs before any
# heavy dependencies so you can see the picture even on a fresh kernel.
import matplotlib.pyplot as _plt
import matplotlib.patches as _patches
import os as _os
```

```
_os.makedirs("images", exist_ok=True)
_fig, _ax = _plt.subplots(figsize=(12, 3.2))
_ax.set_xlim(0, 10); _ax.set_ylim(0, 3); _ax.axis("off")

_boxes = [
    ("Question",  0.2, 1.25, 1.6, 0.9, "#fde68a"),
    ("Retriever", 2.3, 1.25, 1.6, 0.9, "#bfdbfe"),
    ("Passages",  4.4, 1.25, 1.6, 0.9, "#bbf7d0"),
    ("LLM",       6.5, 1.25, 1.6, 0.9, "#ddd6fe"),
    ("Answer",    8.6, 1.25, 1.2, 0.9, "#fecaca"),
    ("Corpus",    2.3, 0.05, 1.6, 0.7, "#e5e7eb"),
]
for text, x, y, w, h, c in _boxes:
    _ax.add_patch(_patches.FancyBboxPatch((x, y), w, h, boxstyle="round,pad=
                                          facecolor=c, edgecolor="#374151",
    _ax.text(x + w / 2, y + h / 2, text, ha="center", va="center", fontsize=

for (x1, x2) in [(1.8, 2.3), (3.9, 4.4), (6.0, 6.5), (8.1, 8.6)]:
    _ax.annotate("", xy=(x2, 1.70), xytext=(x1, 1.70),
                 arrowprops=dict(arrowstyle="->", lw=1.8, color="#374151"))
_ax.annotate("", xy=(3.1, 0.75), xytext=(3.1, 1.25),
             arrowprops=dict(arrowstyle="<->", lw=1.6, color="#374151"))

_ax.set_title("RAG = Retriever + Generator", fontsize=14, fontweight="bold",
_plt.tight_layout()
_fig.savefig("images/what_is_rag.png", dpi=150, bbox_inches="tight")
_plt.show()
```



This simple recipe, often called *naive RAG*, works surprisingly well. But it has plenty of weaknesses: noisy retrieval, coarse chunks, irrelevant context, multi-hop questions, and so on. That is why there are now many RAG variants, each fixing a different problem.

The 8 Architectures We Will Compare

Every architecture in this tutorial uses the same embedding model (`all-MiniLM-L6-v2`), the same FAISS vector store, and the same LLM (`claude-sonnet-4-6`). The only thing that changes is the retrieval / generation logic around them.

#	Architecture	Key Idea
1	Naive RAG	Fixed-size chunking → embed → FAISS → LLM answer (retrieve-then-read baseline)

#	Architecture	Key Idea
2	Advanced RAG	Sentence-aware chunking + multi-query expansion + cross-encoder reranking
3	HyDE	LLM writes a *hypothetical* answer → embed it → retrieve real passages from its neighborhood
4	FLARE	Generate with `[UNCERTAIN: …]` markers → retrieve for uncertain spans → regenerate
5	Self-RAG	LLM decides if retrieval is needed, filters passages (batched), checks grounding
6	CRAG	Classify each retrieved doc Relevant / Ambiguous / Irrelevant (batched); fall back to parametric
7	GraphRAG	Build a co-occurrence knowledge graph, detect communities, retrieve by community summary
8	Agentic RAG	Give Claude `retrieve` and `answer` tools and let it plan multi-step retrievals

We benchmark the first six plus Agentic RAG on **SQuAD span questions** (short factoid answers), then evaluate GraphRAG separately on synthesis questions where it's actually designed to shine, and finally test the retrieve-then-read architectures on a small HotpotQA multi-hop subset.

Installing the Dependencies

The code below installs everything you need to run this notebook. If you are on Colab or a fresh environment, run it once. If you already have these installed locally, feel free to skip it.

`tenacity` is used for retry-with-backoff on API errors; `ipywidgets` silences a tqdm warning when HuggingFace datasets loads.

In [2]:
```
%pip install -q anthropic sentence-transformers faiss-cpu datasets networkx
```

```
Note: you may need to restart the kernel to use updated packages.
```

Import Libraries

A quick note on what each library is for:

- `anthropic` — client for Claude.
- `sentence-transformers` — the embedding model and the cross-encoder used for reranking in Advanced RAG.
- `faiss-cpu` — the vector store.
- `datasets` — SQuAD v1.1 + HotpotQA for the multi-hop section.

- `networkx` + `python-louvain` — the graph and community detection used in GraphRAG.
- `tenacity` — retry with exponential backoff for the API calls.

In [3]:
```
import os
import re
import time
import json
import string
import getpass
import functools
import warnings
from typing import List, Dict, Tuple, Any, Optional
from dataclasses import dataclass, field
from collections import defaultdict, Counter

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import faiss
import networkx as nx
from sentence_transformers import SentenceTransformer, CrossEncoder
from datasets import load_dataset
import anthropic
from tenacity import retry, stop_after_attempt, wait_exponential, retry_if_e

warnings.filterwarnings('ignore')
plt.style.use('seaborn-v0_8-whitegrid')
print("Imports OK")
```

```
Imports OK
```

Setting Up Your Anthropic API Key

All 8 architectures call Claude, so you need an Anthropic API key. You can get one from the Anthropic Console.

Never paste your key directly into a notebook cell. If you do, it can leak via `.ipynb_checkpoints/`, via git history, or when you share the notebook. The cell below loads the key in this order:

1. `ANTHROPIC_API_KEY` already set in the shell you launched Jupyter from.
2. A `.env` file next to the notebook or at the project root, with a line like `ANTHROPIC_API_KEY=sk-ant-...`. The `.gitignore` in this repo already excludes `*.env`, so the file stays local.
3. An interactive `getpass` prompt as a last resort.

Whichever path is used, the key is held in memory only — it is never printed, written into notebook outputs, or persisted by this notebook.

In [4]:
```
def _load_api_key() -> str:
    key = os.environ.get("ANTHROPIC_API_KEY")
    if key:
        return key
    for env_path in (".env", "../.env"):
        if os.path.exists(env_path):
            with open(env_path) as fh:
                for line in fh:
                    line = line.strip()
                    if line.startswith("ANTHROPIC_API_KEY="):
                        return line.split("=", 1)[1].strip().strip('"').stri
    return getpass.getpass("Enter your Anthropic API key (hidden): ")


api_key = _load_api_key()
assert api_key.startswith("sk-"), "That doesn't look like a valid Anthropic
os.environ["ANTHROPIC_API_KEY"] = api_key  # in-memory only

MODEL = "claude-sonnet-4-6"
TOP_K = 5
N_EVAL = 10          # span questions (SQuAD)
N_SYNTHESIS = 5      # synthesis questions for GraphRAG
N_MULTIHOP = 5       # HotpotQA multi-hop questions

client   = anthropic.Anthropic(api_key=api_key)
embedder = SentenceTransformer("all-MiniLM-L6-v2")
reranker = CrossEncoder("cross-encoder/ms-marco-MiniLM-L-6-v2")
print("API key loaded. Models loaded.")
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set
a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights:   0%|          | 0/103 [00:00<?, ?it/s]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architectur
e; not ok if you expect identical arch.
Loading weights:   0%|          | 0/105 [00:00<?, ?it/s]
BertForSequenceClassification LOAD REPORT from: cross-encoder/ms-marco-MiniL
M-L-6-v2
Key                           | Status     |  |
------------------------------+------------+--+-
bert.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architectur
e; not ok if you expect identical arch.
API key loaded. Models loaded.
```

Load the Dataset

For the main benchmark we use SQuAD v1.1 — the Stanford Question Answering Dataset. It pairs Wikipedia paragraphs with questions whose answer is a short span from the paragraph. That makes it a convenient benchmark because:

- The answer is a short, exact string we can score with token F1 *and* an LLM judge.
- Each paragraph can act as one document in our corpus.
- The questions are phrased naturally, so the retriever actually has to work.

The code below downloads the validation split, keeps a ~200-passage corpus, and reserves 10 questions for evaluation.

In [5]:
```
squad = load_dataset("rajpurkar/squad", split="validation")
shuffled = squad.shuffle(seed=42).select(range(200))

eval_questions = []
corpus_passages = []
seen_contexts = set()

for item in shuffled:
    ctx = item["context"]
    if ctx not in seen_contexts:
        corpus_passages.append(ctx)
        seen_contexts.add(ctx)
    if len(eval_questions) < N_EVAL:
        eval_questions.append({
            "question": item["question"],
            "answer":   item["answers"]["text"][0],
            "context":  ctx,
        })

print(f"Corpus: {len(corpus_passages)} passages | Eval: {len(eval_questions)
for i, qa in enumerate(eval_questions):
    print(f"  [{i+1}] Q: {qa['question'][:70]}")
    print(f"       A: {qa['answer']}")
```

```
Corpus: 192 passages | Eval: 10 questions

  [1] Q: In what year did Massachusetts first require children to be educate
d i
       A: 1852
  [2] Q: When were stromules discovered?
       A: 1962
  [3] Q: Which artist who had a major influence on the Gothic Revival is rep
res
       A: Horace Walpole
  [4] Q: In 1890, who did the university decide to team up with?
       A: several regional colleges and universities
  [5] Q: Who got a touchdown making the score 10-7?
       A: Jonathan Stewart
  [6] Q: How many Examination Boards exist in India?
       A: 30
  [7] Q: Who started rumors in 2008 that ABC would sell its ten owned-and-op
era
       A: Caris & Co.
  [8] Q: Which network broadcasted the 50th Super Bowl game?
       A: CBS
  [9] Q: Why was this short termed organization created?
       A: coordinate the response to the embargo
  [10] Q: What does LGM stands for?
       A: Last Glacial Maximum
```

Building the Shared Infrastructure

Before we implement any of the 8 architectures, we set up five pieces of shared infrastructure they all use:

1. **Chunking functions** — how to split a passage into smaller, embeddable pieces.
2. **A `VectorStore` class** — wraps FAISS so each architecture can build an index and search it with one line.
3. **A cached `embed()` helper** — an `lru_cache` around `embedder.encode` so we don't re-embed the same gold answer dozens of times during evaluation.
4. **A retry-wrapped `llm()` helper** — calls Claude with exponential backoff and tracks token usage per architecture.
5. **Evaluation metrics** — token F1, semantic similarity, context recall, and an LLM-as-judge correctness flag.

Chunking

In [6]:
```
def chunk_text(text: str, size: int = 400, overlap: int = 50) -> List[str]:
    """Fixed-size character chunking with overlap."""
    chunks = []
    start = 0
    while start < len(text):
        chunks.append(text[start:start + size])
        start += size - overlap
```

```
    return [c for c in chunks if len(c.strip()) > 20]


def sentence_chunk(text: str, max_chars: int = 500) -> List[str]:
    """Sentence-aware chunking: split on sentence boundaries, pack to ~max_c
    sentences = re.split(r'(?<=[.!?])\s+', text)
    chunks, current = [], ""
    for sent in sentences:
        if len(current) + len(sent) + 1 <= max_chars:
            current = (current + " " + sent).strip()
        else:
            if current:
                chunks.append(current)
            current = sent
    if current:
        chunks.append(current)
    return [c for c in chunks if len(c.strip()) > 20]
```

Cached Embeddings

During evaluation the same gold answers and questions get embedded repeatedly (for semantic similarity, for context recall, etc.). Wrapping the encoder in an `lru_cache` avoids re-computing those embeddings.

In [7]:
```
@functools.lru_cache(maxsize=4096)
def _embed_cached(text: str) -> tuple:
    """Cached single-string embedding. Returns a tuple so it's hashable if e
    return tuple(embedder.encode([text], normalize_embeddings=True)[0].tolis


def embed_one(text: str) -> np.ndarray:
    return np.array(_embed_cached(text), dtype="float32")


def embed_many(texts: List[str]) -> np.ndarray:
    return np.stack([embed_one(t) for t in texts]).astype("float32")
```

Vector Store

Uses FAISS `IndexFlatIP` because we L2-normalize the embeddings, making inner-product equivalent to cosine similarity.

In [8]:
```
@dataclass
class VectorStore:
    chunks: List[str] = field(default_factory=list)
    index: Any = field(default=None, repr=False)

    def build(self, passages: List[str], chunk_fn=chunk_text):
        self.chunks = []
        for p in passages:
            self.chunks.extend(chunk_fn(p))
        print(f"  Building index over {len(self.chunks)} chunks…", end=" ")
```

```
        embs = embedder.encode(self.chunks, batch_size=64, show_progress_bar
                               normalize_embeddings=True).astype("float32")
        self.index = faiss.IndexFlatIP(embs.shape[1])
        self.index.add(embs)
        print("done.")

    def search(self, query: str, k: int = 5) -> List[Tuple[str, float]]:
        q_emb = embedder.encode([query], normalize_embeddings=True).astype("
        scores, idxs = self.index.search(q_emb, min(k, len(self.chunks)))
        return [(self.chunks[i], float(s)) for i, s in zip(idxs[0], scores[0
```

LLM Wrapper with Retry + Token Tracking

Three things to notice here:

1. **Retry with exponential backoff** — one 429 or 529 no longer ruins an architecture's score for a question.
2. **A global token counter** (`TOKEN_USAGE`) we tag with the architecture name, so we can report cost at the end.
3. **Span-style system prompt** — we tell Claude to answer with the shortest literal span from the context rather than writing prose. This is what actually fixes the misleadingly-low F1 scores you see on SQuAD-style benchmarks.

In [9]:
```
TOKEN_USAGE: Dict[str, Dict[str, int]] = defaultdict(lambda: {"input": 0, "c
CURRENT_ARCH: List[str] = ["misc"]  # a one-element list we can mutate from


class _ArchScope:
    """with arch('Naive RAG'):  all llm() calls inside get tagged to that ar
    def __init__(self, name): self.name = name
    def __enter__(self): CURRENT_ARCH.append(self.name); return self
    def __exit__(self, *a): CURRENT_ARCH.pop()


def arch(name: str) -> _ArchScope:
    return _ArchScope(name)


SPAN_SYSTEM = (
    "You are a precise question-answering assistant. "
    "When given a context, answer with the shortest literal span from the co
    "No prose, no punctuation beyond what is in the span, no explanations. "
    "If the answer is not in the context, reply exactly: NOT_FOUND."
)


@retry(
    stop=stop_after_attempt(4),
    wait=wait_exponential(multiplier=1, min=1, max=20),
    retry=retry_if_exception_type((anthropic.RateLimitError, anthropic.APICo
    reraise=True,
)
```

```
def _call_llm(prompt: str, system: str, max_tokens: int):
    return client.messages.create(
        model=MODEL, max_tokens=max_tokens, system=system,
        messages=[{"role": "user", "content": prompt}],
    )


def llm(prompt: str, system: str = SPAN_SYSTEM, max_tokens: int = 128) -> st
    """Call Claude with retry + token tracking. Returns plain text."""
    msg = _call_llm(prompt, system, max_tokens)
    usage = getattr(msg, "usage", None)
    if usage is not None:
        bucket = TOKEN_USAGE[CURRENT_ARCH[-1]]
        bucket["input"]  += getattr(usage, "input_tokens", 0)
        bucket["output"] += getattr(usage, "output_tokens", 0)
        bucket["calls"]  += 1
    return msg.content[0].text.strip()
```

Evaluation Metrics

We score each architecture on five things:

- **Answer F1** — token-level F1 between predicted and gold answer (same metric SQuAD uses). Good continuity with prior work but punishes prose answers.
- **Answer Correctness (LLM judge)** — one extra Claude call per answer that returns CORRECT / INCORRECT. Forgiving of paraphrase and formatting. This is the metric we'll foreground.
- **Semantic Similarity** — cosine similarity between predicted and gold answer embeddings.
- **Context Recall** — max cosine similarity between gold answer and any retrieved chunk. Tells us whether the retriever even found the right passage.
- **Latency** — wall-clock seconds per question.

In [10]:
```
def normalize_answer(s: str) -> str:
    s = s.lower()
    s = re.sub(r'\b(a|an|the)\b', ' ', s)
    s = s.translate(str.maketrans('', '', string.punctuation))
    return ' '.join(s.split())


def f1_score(pred: str, gold: str) -> float:
    pred_toks = normalize_answer(pred).split()
    gold_toks = normalize_answer(gold).split()
    common = Counter(pred_toks) & Counter(gold_toks)
    num_same = sum(common.values())
    if num_same == 0:
        return 0.0
    precision = num_same / len(pred_toks)
    recall    = num_same / len(gold_toks)
    return 2 * precision * recall / (precision + recall)
```

```
def semantic_similarity(pred: str, gold: str) -> float:
    embs = np.stack([embed_one(pred), embed_one(gold)])
    return float(np.dot(embs[0], embs[1]))


def context_recall(gold_answer: str, contexts: List[str]) -> float:
    if not contexts:
        return 0.0
    gold_emb = embed_one(gold_answer)
    ctx_embs = embed_many(contexts)
    return float(np.max(ctx_embs @ gold_emb))


JUDGE_SYSTEM = (
    "You grade question-answering systems. Given a question, a gold answer,
    "decide whether the prediction is CORRECT — a faithful paraphrase or sup
    "or INCORRECT. Reply with exactly one token: CORRECT or INCORRECT."
)


def llm_judge(question: str, pred: str, gold: str) -> int:
    """Return 1 if the predicted answer is judged correct, 0 otherwise."""
    with arch("judge"):
        verdict = llm(
            f"Question: {question}\nGold answer: {gold}\nPredicted answer: {
            system=JUDGE_SYSTEM, max_tokens=4,
        )
    return 1 if "CORRECT" in verdict.upper() and "INCORRECT" not in verdict.


@dataclass
class EvalResult:
    architecture: str
    question:     str
    answer:       str
    gold_answer:  str
    contexts:     List[str]
    latency:      float
    f1:           float = 0.0
    correct:      int   = 0
    semantic_sim: float = 0.0
    ctx_recall:   float = 0.0

    def compute_metrics(self) -> "EvalResult":
        self.f1           = f1_score(self.answer, self.gold_answer)
        self.correct      = llm_judge(self.question, self.answer, self.gold_
        self.semantic_sim = semantic_similarity(self.answer, self.gold_answe
        self.ctx_recall   = context_recall(self.gold_answer, self.contexts)
        return self
```

Build the Vector Stores

Naive RAG uses fixed-size chunks; Advanced RAG uses sentence-aware chunks.

In [11]:
```
print("Building naive_store (fixed-size chunks)…")
naive_store = VectorStore()
naive_store.build(corpus_passages, chunk_fn=chunk_text)

print("Building advanced_store (sentence-aware chunks)…")
advanced_store = VectorStore()
advanced_store.build(corpus_passages, chunk_fn=sentence_chunk)

print("Vector stores ready.")
```

```
Building naive_store (fixed-size chunks)…
  Building index over 511 chunks… done.
Building advanced_store (sentence-aware chunks)…
  Building index over 436 chunks… done.
Vector stores ready.
```

A shared "answer from context" helper

All of the retrieve-then-read architectures end with the same pattern: hand the LLM the retrieved chunks and ask for the answer. The helper below centralizes that so we can tweak the prompt in one place.

In [12]:
```
ANSWER_USER_TEMPLATE = (
    "Context:\n{ctx}\n\n"
    "Question: {q}\n\n"
    "Answer with the shortest literal span from the context. No prose."
)


def answer_from_context(question: str, contexts: List[str]) -> str:
    if not contexts:
        return llm(f"Question: {question}\nAnswer:", max_tokens=32)
    ctx_str = "\n\n".join(f"[{i+1}] {c}" for i, c in enumerate(contexts))
    return llm(ANSWER_USER_TEMPLATE.format(ctx=ctx_str, q=question), max_tok
```

1. Naive RAG

Step 1: Chunk the corpus into fixed-size windows (400 chars, 50-char overlap).

Step 2: Embed chunks with `all-MiniLM-L6-v2` and store in FAISS.

Step 3: At query time, retrieve the top-K most similar chunks.

Step 4: Concatenate those chunks as context and ask the LLM to answer.

This is the retrieve-then-read baseline. Every other architecture in this tutorial can be understood as a modification of one of these four steps.

In [13]:
```
def naive_rag(question: str, store: "VectorStore" = None,
              arch_name: str = "Naive RAG") -> Tuple[str, List[str], float]:
    store = store or naive_store
```

```
    t0 = time.time()
    with arch(arch_name):
        contexts = [r[0] for r in store.search(question, k=TOP_K)]
        answer   = answer_from_context(question, contexts)
    return answer, contexts, time.time() - t0


_ans, _ctx, _lat = naive_rag(eval_questions[0]["question"])
print(f"Naive RAG smoke test ({_lat:.2f}s): {_ans[:120]}")
```

```
Naive RAG smoke test (1.85s): 1852
```

2. Advanced RAG

Three upgrades on the naive baseline:

1. **Sentence-aware chunking** — chunks respect sentence boundaries (~500 chars).
2. **Multi-query expansion** — the LLM rephrases the question two additional ways.
3. **Cross-encoder reranking** — all candidates are re-scored with `ms-marco-MiniLM-L-6-v2`, a pointwise relevance model.

In [14]:
```
REPHRASE_SYSTEM = "You rephrase questions. Return exactly 3 rephrasings, one


def advanced_rag(question: str, store: "VectorStore" = None,
                 arch_name: str = "Advanced RAG") -> Tuple[str, List[str], f
    store = store or advanced_store
    t0 = time.time()
    with arch(arch_name):
        raw = llm(f"Rephrase: {question}", system=REPHRASE_SYSTEM, max_toker
        queries = [question] + [q.strip() for q in raw.strip().split("\n") i

        seen, candidates = set(), []
        for q in queries:
            for chunk, _ in store.search(q, k=3):
                if chunk not in seen:
                    candidates.append(chunk); seen.add(chunk)

        if candidates:
            scores = reranker.predict([(question, c) for c in candidates])
            ranked = sorted(zip(scores, candidates), reverse=True)
            contexts = [c for _, c in ranked[:TOP_K]]
        else:
            contexts = []

        answer = answer_from_context(question, contexts)
    return answer, contexts, time.time() - t0


_ans, _ctx, _lat = advanced_rag(eval_questions[0]["question"])
print(f"Advanced RAG smoke test ({_lat:.2f}s): {_ans[:120]}")
```

```
Advanced RAG smoke test (4.01s): 1852
```

3. HyDE (Hypothetical Document Embeddings)

Key insight: the embedding space aligns better if we search using a *hypothetical answer* rather than the raw question.

> Gao et al., 2022 — *Precise Zero-Shot Dense Retrieval without Relevance Labels*

In [15]:
```
HYPO_SYSTEM = "You write a brief, confident factual answer to the user's que


def hyde_rag(question: str) -> Tuple[str, List[str], float]:
    t0 = time.time()
    with arch("HyDE"):
        hypo_answer = llm(question, system=HYPO_SYSTEM, max_tokens=60)

        hypo_emb = embedder.encode([hypo_answer], normalize_embeddings=True)
        _, idxs = naive_store.index.search(hypo_emb, TOP_K)
        contexts = [naive_store.chunks[i] for i in idxs[0] if i >= 0]

        answer = answer_from_context(question, contexts)
    return answer, contexts, time.time() - t0


_ans, _ctx, _lat = hyde_rag(eval_questions[0]["question"])
print(f"HyDE smoke test ({_lat:.2f}s): {_ans[:120]}")
```

```
HyDE smoke test (2.81s): 1852
```

4. FLARE (Forward-Looking Active REtrieval)

Key insight: retrieve *on demand* whenever the model is uncertain, rather than once upfront.

> Jiang et al., 2023 — *Active Retrieval Augmented Generation*

In [16]:
```
FLARE_DRAFT_SYSTEM = (
    "Answer the user's question. Wrap any part you are uncertain about in [U
    "Keep the answer to one sentence."
)


def flare_rag(question: str) -> Tuple[str, List[str], float]:
    t0 = time.time()
    with arch("FLARE"):
        draft = llm(question, system=FLARE_DRAFT_SYSTEM, max_tokens=160)
        uncertain_spans = re.findall(r'\[UNCERTAIN:\s*(.*?)\]', draft, re.DO

        all_contexts, seen = [], set()
        for chunk, _ in naive_store.search(question, k=3):
```

```
            if chunk not in seen:
                all_contexts.append(chunk); seen.add(chunk)
        for span in uncertain_spans[:3]:
            for chunk, _ in naive_store.search(f"{question} {span}", k=2):
                if chunk not in seen:
                    all_contexts.append(chunk); seen.add(chunk)

        contexts = all_contexts[:TOP_K]
        answer = answer_from_context(question, contexts)
    return answer, contexts, time.time() - t0


_ans, _ctx, _lat = flare_rag(eval_questions[0]["question"])
print(f"FLARE smoke test ({_lat:.2f}s): {_ans[:120]}")
```

```
FLARE smoke test (2.26s): 1852
```

5. Self-RAG (with batched relevance filtering)

Key insight: the LLM should be an *active participant* in the retrieval process — deciding when to retrieve, which passages are actually relevant, and whether its final answer is grounded.

The original Self-RAG does *one LLM call per retrieved passage* to score relevance. On 7 candidates that's 7 sequential API calls per question. We fix this by **batching** all candidates into a single call that returns a CSV of 1/0 labels.

> Asai et al., 2023 — *Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection*

In [17]:
```
RELEVANCE_BATCH_SYSTEM = (
    "You score passage relevance. Reply with ONLY a comma-separated list of
    "1 = useful for answering the question, 0 = not useful. No explanation."
)


def _batched_relevance(question: str, candidates: List[str]) -> List[bool]:
    numbered = "\n".join(f"[{i+1}] {c[:350]}" for i, c in enumerate(candidat
    reply = llm(
        f"Question: {question}\n\nPassages:\n{numbered}\n\nCSV of 1/0 (lengt
        system=RELEVANCE_BATCH_SYSTEM, max_tokens=64,
    )
    labels = re.findall(r"[01]", reply)
    labels = (labels + ["0"] * len(candidates))[: len(candidates)]
    return [lab == "1" for lab in labels]


def self_rag(question: str) -> Tuple[str, List[str], float]:
    t0 = time.time()
    with arch("Self-RAG"):
        retrieval_decision = llm(
            f"Can you answer confidently without external sources? Reply YES
```

```
            system="Reply with exactly YES or NO.", max_tokens=4,
        )
        needs_retrieval = "NO" in retrieval_decision.upper()

        contexts = []
        if needs_retrieval:
            candidates = [r[0] for r in naive_store.search(question, k=TOP_K
            keep = _batched_relevance(question, candidates)
            contexts = [c for c, k in zip(candidates, keep) if k][:TOP_K] or

        answer = answer_from_context(question, contexts)
    return answer, contexts, time.time() - t0


_ans, _ctx, _lat = self_rag(eval_questions[0]["question"])
print(f"Self-RAG smoke test ({_lat:.2f}s): {_ans[:120]}")
```

```
Self-RAG smoke test (3.08s): 1852
```

6. Corrective RAG (CRAG, with batched classification)

Key insight: not all retrieved documents are equally useful. CRAG explicitly classifies each retrieved document as Relevant, Ambiguous, or Irrelevant — and falls back to parametric knowledge if nothing is relevant. We batch the classification into a single call.

> Yan et al., 2024 — *Corrective Retrieval Augmented Generation*

In [18]:

```
CRAG_BATCH_SYSTEM = (
    "You classify passage relevance. For each numbered passage, output a sir
    "R (relevant), A (ambiguous), or I (irrelevant). Reply with a comma-sepa
)


def _batched_classify(question: str, candidates: List[str]) -> List[str]:
    numbered = "\n".join(f"[{i+1}] {c[:400]}" for i, c in enumerate(candidat
    reply = llm(
        f"Question: {question}\n\nPassages:\n{numbered}\n\nCSV of R/A/I (ler
        system=CRAG_BATCH_SYSTEM, max_tokens=64,
    )
    letters = re.findall(r"[RAIrai]", reply)
    letters = [c.upper() for c in letters]
    letters = (letters + ["I"] * len(candidates))[: len(candidates)]
    return letters


def crag_rag(question: str) -> Tuple[str, List[str], float]:
    t0 = time.time()
    with arch("CRAG"):
        candidates = [r[0] for r in naive_store.search(question, k=TOP_K)]
        labels = _batched_classify(question, candidates)

        relevant, ambiguous = [], []
        for chunk, lab in zip(candidates, labels):
```

```
            if lab == "R":
                relevant.append(chunk)
            elif lab == "A":
                ambiguous.append(chunk)

        final_contexts = list(relevant)
        for chunk in ambiguous:
            extracted = llm(
                f"Question: {question}\nPassage: {chunk[:400]}\n"
                f"Return only the sentences from the passage that directly a
                f"Reply NONE if none do.",
                system="Extract relevant sentences verbatim or reply NONE.",
            )
            if extracted.strip().upper() != "NONE" and len(extracted.strip()
                final_contexts.append(extracted.strip())

        answer = answer_from_context(question, final_contexts)
    return answer, final_contexts, time.time() - t0


_ans, _ctx, _lat = crag_rag(eval_questions[0]["question"])
print(f"CRAG smoke test ({_lat:.2f}s): {_ans[:120]}")
```

```
CRAG smoke test (1.78s): 1852
```

7. Agentic RAG

Give Claude two tools (`retrieve` , `answer`) and let it plan its own retrieval steps. This is where multi-step reasoning pays off — see the HotpotQA section later.

In [19]:

```
AGENTIC_TOOLS = [
    {
        "name": "retrieve",
        "description": "Search the corpus for passages. Returns top-5 chunks
        "input_schema": {
            "type": "object",
            "properties": {"query": {"type": "string", "description": "Searc
            "required": ["query"],
        },
    },
    {
        "name": "answer",
        "description": "Provide the final answer. Return the shortest litera
        "input_schema": {
            "type": "object",
            "properties": {"answer": {"type": "string", "description": "Shor
            "required": ["answer"],
        },
    },
]

AGENT_SYSTEM = (
    "You are a question-answering agent. Use the 'retrieve' tool one or more
    "then call the 'answer' tool with the shortest literal span from those p
```

```
    "No prose, no explanations in the answer. Do not answer from prior knowl
)


@retry(stop=stop_after_attempt(4), wait=wait_exponential(multiplier=1, min=1
       retry=retry_if_exception_type((anthropic.RateLimitError, anthropic.AP
                                      anthropic.InternalServerError)), reraiı
def _agent_step(messages):
    return client.messages.create(model=MODEL, max_tokens=400, system=AGENT_
                                  tools=AGENTIC_TOOLS, messages=messages)


def agentic_rag(question: str, store: "VectorStore" = None,
                arch_name: str = "Agentic RAG") -> Tuple[str, List[str], flo
    store = store or naive_store
    t0 = time.time()
    with arch(arch_name):
        messages = [{"role": "user", "content": f"Question: {question}"}]
        all_contexts, final_answer = [], ""
        for _ in range(4):  # max 4 iterations
            response = _agent_step(messages)
            # token tracking (this path bypasses llm() so we tally manually)
            usage = getattr(response, "usage", None)
            if usage is not None:
                bucket = TOKEN_USAGE[CURRENT_ARCH[-1]]
                bucket["input"]  += getattr(usage, "input_tokens", 0)
                bucket["output"] += getattr(usage, "output_tokens", 0)
                bucket["calls"]  += 1

            messages.append({"role": "assistant", "content": response.conter
            if response.stop_reason == "end_turn":
                for block in response.content:
                    if hasattr(block, "text"):
                        final_answer = block.text.strip()
                break
            if response.stop_reason != "tool_use":
                break

            tool_results, done = [], False
            for block in response.content:
                if block.type != "tool_use":
                    continue
                if block.name == "retrieve":
                    chunks = [r[0] for r in store.search(block.input.get("qu
                    for c in chunks:
                        if c not in all_contexts:
                            all_contexts.append(c)
                    tool_results.append({"type": "tool_result", "tool_use_ic
                                         "content": "\n\n".join(f"[{i+1}] {c
                elif block.name == "answer":
                    final_answer = block.input.get("answer", "")
                    tool_results.append({"type": "tool_result", "tool_use_ic
                                         "content": "Recorded."})
                    done = True
            if tool_results:
                messages.append({"role": "user", "content": tool_results})
```

```
            if done:
                break

        final_answer = final_answer or "NOT_FOUND"
    return final_answer, all_contexts[:TOP_K], time.time() - t0


_ans, _ctx, _lat = agentic_rag(eval_questions[0]["question"])
print(f"Agentic RAG smoke test ({_lat:.2f}s): {_ans[:120]}")
```

```
Agentic RAG smoke test (3.22s): 1852
```

Running the Main Benchmark

Seven architectures (GraphRAG is evaluated separately on synthesis questions below) on the same 10 SQuAD questions. For each `(architecture, question)` pair we compute F1, LLM-judge correctness, semantic similarity, context recall, and latency.

In [20]:
```
MAIN_ARCHITECTURES = {
    "Naive RAG":    naive_rag,
    "Advanced RAG": advanced_rag,
    "HyDE":         hyde_rag,
    "FLARE":        flare_rag,
    "Self-RAG":     self_rag,
    "CRAG":         crag_rag,
    "Agentic RAG":  agentic_rag,
}

main_results: Dict[str, List[EvalResult]] = {name: [] for name in MAIN_ARCHI

for idx, qa in enumerate(eval_questions):
    print(f"\n{'='*60}\nQ{idx+1}: {qa['question']}\nGold: {qa['answer']}")
    for arch_name, fn in MAIN_ARCHITECTURES.items():
        try:
            ans, ctxs, lat = fn(qa["question"])
            r = EvalResult(arch_name, qa["question"], ans, qa["answer"], ctx
            r.compute_metrics()
            main_results[arch_name].append(r)
            print(f"  {arch_name:13s} | F1={r.f1:.3f} Correct={r.correct} "
                  f"Sem={r.semantic_sim:.3f} CtxR={r.ctx_recall:.3f} {lat:5.
        except Exception as e:
            print(f"  {arch_name:13s} | ERROR: {e}")
        time.sleep(0.2)

print("\nMain benchmark complete.")
```

```
============================================================
Q1: In what year did Massachusetts first require children to be educated in
schools?
Gold: 1852
  Naive RAG     | F1=1.000 Correct=1 Sem=1.000 CtxR=0.211   0.9s  | 1852
  Advanced RAG  | F1=1.000 Correct=1 Sem=1.000 CtxR=0.251   5.5s  | 1852
  HyDE          | F1=1.000 Correct=1 Sem=1.000 CtxR=0.382   2.4s  | 1852
  FLARE         | F1=1.000 Correct=1 Sem=1.000 CtxR=0.211   3.4s  | 1852
  Self-RAG      | F1=1.000 Correct=1 Sem=1.000 CtxR=0.211   3.2s  | 1852
  CRAG          | F1=1.000 Correct=1 Sem=1.000 CtxR=0.211   2.2s  | 1852
  Agentic RAG   | F1=1.000 Correct=1 Sem=1.000 CtxR=0.211   3.2s  | 1852

============================================================
Q2: When were stromules discovered?
Gold: 1962
  Naive RAG     | F1=1.000 Correct=1 Sem=1.000 CtxR=0.085   0.9s  | 1962
  Advanced RAG  | F1=1.000 Correct=1 Sem=1.000 CtxR=0.117   4.1s  | 1962
  HyDE          | F1=1.000 Correct=1 Sem=1.000 CtxR=0.173   3.2s  | 1962
  FLARE         | F1=1.000 Correct=1 Sem=1.000 CtxR=0.085   3.1s  | 1962
  Self-RAG      | F1=1.000 Correct=1 Sem=1.000 CtxR=0.020   3.3s  | 1962
  CRAG          | F1=1.000 Correct=1 Sem=1.000 CtxR=0.020   2.1s  | 1962
  Agentic RAG   | F1=1.000 Correct=1 Sem=1.000 CtxR=0.020   3.7s  | 1962

============================================================
Q3: Which artist who had a major influence on the Gothic Revival is represen
ted in the V&A's British galleries?
Gold: Horace Walpole
  Naive RAG     | F1=1.000 Correct=1 Sem=1.000 CtxR=0.453   0.9s  | Horace W
alpole
  Advanced RAG  | F1=1.000 Correct=1 Sem=1.000 CtxR=0.424   4.1s  | Horace W
alpole
  HyDE          | F1=1.000 Correct=1 Sem=1.000 CtxR=0.453   2.1s  | Horace W
alpole
  FLARE         | F1=1.000 Correct=1 Sem=1.000 CtxR=0.453   2.2s  | Horace W
alpole
  Self-RAG      | F1=1.000 Correct=1 Sem=1.000 CtxR=0.453   3.6s  | Horace W
alpole
  CRAG          | F1=1.000 Correct=1 Sem=1.000 CtxR=0.453   2.3s  | Horace W
alpole
  Agentic RAG   | F1=1.000 Correct=1 Sem=1.000 CtxR=0.453   2.7s  | Horace W
alpole

============================================================
Q4: In 1890, who did the university decide to team up with?
Gold: several regional colleges and universities
  Naive RAG     | F1=0.133 Correct=1 Sem=0.594 CtxR=0.525   1.5s  | Des Moin
es College, Kalamazoo College, Butler Univ
  Advanced RAG  | F1=0.133 Correct=1 Sem=0.594 CtxR=0.546   2.8s  | Des Moin
es College, Kalamazoo College, Butler Univ
  HyDE          | F1=0.500 Correct=1 Sem=0.685 CtxR=0.525   2.8s  | several
regional colleges and universities: Des Mo
  FLARE         | F1=0.133 Correct=1 Sem=0.594 CtxR=0.525   2.8s  | Des Moin
es College, Kalamazoo College, Butler Univ
  Self-RAG      | F1=0.133 Correct=1 Sem=0.594 CtxR=0.525   3.3s  | Des Moin
es College, Kalamazoo College, Butler Univ
  CRAG          | F1=0.133 Correct=1 Sem=0.594 CtxR=0.525   5.3s  | Des Moin
```

```
es College, Kalamazoo College, Butler Univ
  Agentic RAG   | F1=0.133 Correct=1 Sem=0.594 CtxR=0.525   6.1s  | Des Moin
es College, Kalamazoo College, Butler Univ

============================================================
Q5: Who got a touchdown making the score 10-7?
Gold: Jonathan Stewart
  Naive RAG     | F1=1.000 Correct=1 Sem=1.000 CtxR=0.307   1.2s  | Jonathan
Stewart
  Advanced RAG  | F1=1.000 Correct=1 Sem=1.000 CtxR=0.337   2.5s  | Jonathan
Stewart
  HyDE          | F1=0.000 Correct=0 Sem=0.100 CtxR=0.154   2.4s  | NOT_FOUN
D
  FLARE         | F1=1.000 Correct=1 Sem=1.000 CtxR=0.307   3.4s  | Jonathan
Stewart
  Self-RAG      | F1=1.000 Correct=1 Sem=1.000 CtxR=0.307   3.1s  | Jonathan
Stewart
  CRAG          | F1=1.000 Correct=1 Sem=1.000 CtxR=0.307   2.6s  | Jonathan
Stewart
  Agentic RAG   | F1=1.000 Correct=1 Sem=1.000 CtxR=0.307   2.5s  | Jonathan
Stewart

============================================================
Q6: How many Examination Boards exist in India?
Gold: 30
  Naive RAG     | F1=1.000 Correct=1 Sem=1.000 CtxR=0.090   1.4s  | 30
  Advanced RAG  | F1=1.000 Correct=1 Sem=1.000 CtxR=0.125   2.2s  | 30
  HyDE          | F1=1.000 Correct=1 Sem=1.000 CtxR=0.108   2.1s  | 30
  FLARE         | F1=1.000 Correct=1 Sem=1.000 CtxR=0.090   3.4s  | 30
  Self-RAG      | F1=1.000 Correct=1 Sem=1.000 CtxR=0.014   3.1s  | 30
  CRAG          | F1=1.000 Correct=1 Sem=1.000 CtxR=0.014   1.9s  | 30
  Agentic RAG   | F1=1.000 Correct=1 Sem=1.000 CtxR=0.066   3.0s  | 30

============================================================
Q7: Who started rumors in 2008 that ABC would sell its ten owned-and-operate
d stations?
Gold: Caris & Co.
  Naive RAG     | F1=0.000 Correct=0 Sem=0.087 CtxR=0.149   0.8s  | NOT_FOUN
D
  Advanced RAG  | F1=0.000 Correct=0 Sem=0.087 CtxR=0.141   3.9s  | NOT_FOUN
D
  HyDE          | F1=0.000 Correct=0 Sem=0.087 CtxR=0.226   1.5s  | NOT_FOUN
D
  FLARE         | F1=0.000 Correct=0 Sem=0.087 CtxR=0.149   3.2s  | NOT_FOUN
D
  Self-RAG      | F1=0.000 Correct=0 Sem=0.087 CtxR=0.149   2.3s  | NOT_FOUN
D
  CRAG          | F1=0.000 Correct=0 Sem=0.087 CtxR=0.000   2.1s  | NOT_FOUN
D
  Agentic RAG   | F1=0.000 Correct=0 Sem=0.087 CtxR=0.149   5.8s  | NOT_FOUN
D

============================================================
Q8: Which network broadcasted the 50th Super Bowl game?
Gold: CBS
  Naive RAG     | F1=1.000 Correct=1 Sem=1.000 CtxR=0.392   0.8s  | CBS
```

```
  Advanced RAG  | F1=1.000 Correct=1 Sem=1.000 CtxR=0.376   2.9s  | CBS
  HyDE          | F1=1.000 Correct=1 Sem=1.000 CtxR=0.367   2.4s  | CBS
  FLARE         | F1=1.000 Correct=1 Sem=1.000 CtxR=0.318   4.0s  | CBS
  Self-RAG      | F1=0.000 Correct=0 Sem=0.103 CtxR=0.000   1.8s  | NOT_FOUN
D
  CRAG          | F1=1.000 Correct=1 Sem=1.000 CtxR=0.413   3.9s  | CBS
  Agentic RAG   | F1=1.000 Correct=1 Sem=1.000 CtxR=0.392   3.4s  | CBS

============================================================
Q9: Why was this short termed organization created?
Gold: coordinate the response to the embargo
  Naive RAG     | F1=0.000 Correct=0 Sem=0.020 CtxR=0.012   1.6s  | managing
the Apollo program would exceed the capab
  Advanced RAG  | F1=0.000 Correct=0 Sem=0.020 CtxR=0.057   3.7s  | NOT_FOUN
D
  HyDE          | F1=0.000 Correct=0 Sem=0.020 CtxR=0.063   3.3s  | managing
the Apollo program would exceed the capab
  FLARE         | F1=0.000 Correct=0 Sem=0.020 CtxR=0.012   4.8s  | managing
the Apollo program would exceed the capab
  Self-RAG      | F1=0.000 Correct=0 Sem=0.020 CtxR=-0.050   4.3s  | managin
g the Apollo program would exceed the capab
  CRAG          | F1=0.000 Correct=0 Sem=0.020 CtxR=-0.050   2.1s  | managin
g the Apollo program would exceed the capab
  Agentic RAG   | F1=0.040 Correct=0 Sem=-0.031 CtxR=0.069   5.6s  | I was
n't able to find specific information about t

============================================================
Q10: What does LGM stands for?
Gold: Last Glacial Maximum
  Naive RAG     | F1=0.000 Correct=0 Sem=0.078 CtxR=0.082   1.2s  | NOT_FOUN
D
  Advanced RAG  | F1=0.000 Correct=0 Sem=0.078 CtxR=0.073   3.4s  | NOT_FOUN
D
  HyDE          | F1=1.000 Correct=1 Sem=1.000 CtxR=0.472   2.5s  | Last Gla
cial Maximum
  FLARE         | F1=0.000 Correct=0 Sem=0.078 CtxR=0.082   3.6s  | NOT_FOUN
D
  Self-RAG      | F1=0.000 Correct=0 Sem=0.078 CtxR=0.082   4.5s  | NOT_FOUN
D
  CRAG          | F1=0.000 Correct=0 Sem=0.078 CtxR=0.000   2.4s  | NOT_FOUN
D
  Agentic RAG   | F1=1.000 Correct=1 Sem=1.000 CtxR=0.082   6.5s  | Last Gla
cial Maximum

Main benchmark complete.
```

Aggregated Results (Main Benchmark)

The table below averages each metric across the 10 questions. **Answer Correctness** (LLM-judge) is the headline metric — it forgives paraphrasing that token F1 punishes.

In [21]:
```
def aggregate(results_dict):
    rows = []
    for arch_name, results in results_dict.items():
        if not results:
```

```
            continue
        toks = TOKEN_USAGE.get(arch_name, {"input": 0, "output": 0, "calls":
        rows.append({
            "Architecture":        arch_name,
            "Correctness":         round(np.mean([r.correct      for r in re
            "Answer F1":           round(np.mean([r.f1           for r in re
            "Semantic Similarity": round(np.mean([r.semantic_sim for r in re
            "Context Recall":      round(np.mean([r.ctx_recall   for r in re
            "Avg Latency (s)":     round(np.mean([r.latency      for r in re
            "In Tokens":           toks["input"],
            "Out Tokens":          toks["output"],
            "n":                   len(results),
        })
    return pd.DataFrame(rows).set_index("Architecture")


df_main = aggregate(main_results)
df_main
```

Out[21]:

Architecture	Correctness	Answer F1	Semantic Similarity	Context Recall	Avg Latency (s)	In Tokens	Out Tokens	n
Naive RAG	0.7	0.613	0.678	0.230	1.11	5537	104	10
Advanced RAG	0.7	0.613	0.678	0.245	3.50	5345	609	10
HyDE	0.7	0.650	0.689	0.292	2.47	5937	377	10
FLARE	0.7	0.613	0.678	0.223	3.39	4417	526	10
Self-RAG	0.6	0.513	0.588	0.171	3.25	9342	321	10
CRAG	0.7	0.613	0.678	0.189	2.68	7869	270	10
Agentic RAG	0.8	0.717	0.765	0.227	4.25	31165	1654	10

GraphRAG on Synthesis Questions

GraphRAG retrieves *paraphrased community summaries*, not literal passages — so it structurally cannot score well on SQuAD-style span questions ("What year did X happen?"). It is designed for **synthesis questions** where the answer is a multi-sentence summary across many passages.

For synthesis there is no meaningful gold span to score against, so instead of making up target answers and grading them with an LLM judge (which would just reward GraphRAG for being plausible), we evaluate **qualitatively**: we ask five synthesis-style questions, print the answers, and let the reader judge. Latency is still useful to track.

> Edge et al., 2024 — *From Local to Global: A Graph RAG Approach to Query-Focused Summarization*

In [22]:
```
class GraphRAGSystem:
    def __init__(self):
        self.graph = None
        self.communities = {}
        self.community_summaries: List[Tuple[int, str]] = []
        self.summary_embeddings = None

    def build_graph(self, passages: List[str]):
        self.graph = nx.Graph()
        edge_weights: Dict[Tuple[str, str], int] = defaultdict(int)
        entity_pattern = re.compile(r'\b([A-Z][a-z]+(?: [A-Z][a-z]+){0,3})\b
        for passage in passages:
            entities = list(set(entity_pattern.findall(passage)))
            for i in range(len(entities)):
                for j in range(i + 1, len(entities)):
                    e1, e2 = sorted([entities[i], entities[j]])
                    edge_weights[(e1, e2)] += 1
        for (e1, e2), w in edge_weights.items():
            self.graph.add_edge(e1, e2, weight=w)
        print(f"  Graph: {self.graph.number_of_nodes()} nodes, {self.graph.r

    def detect_communities(self):
        try:
            import community as community_louvain
            self.communities = community_louvain.best_partition(self.graph)
        except Exception:
            nodes = list(self.graph.nodes())
            self.communities = {n: i // 20 for i, n in enumerate(nodes)}
        print(f"  Communities: {len(set(self.communities.values()))}")

    def summarize_communities(self, passages: List[str], top_n: int = 8):
        community_nodes: Dict[int, List[str]] = defaultdict(list)
        for node, cid in self.communities.items():
            community_nodes[cid].append(node)
        top = sorted(community_nodes.items(), key=lambda x: len(x[1]), rever

        with arch("GraphRAG"):
            for cid, nodes in top:
                node_set = set(nodes)
                scored = [(sum(1 for n in node_set if n in p), p) for p in p
                scored = sorted([(s, p) for s, p in scored if s > 0], revers
                top_passages = [p for _, p in scored[:3]] or [" ".join(nodes
                ctx = "\n".join(p[:300] for p in top_passages)
                summary = llm(
                    f"Entities: {', '.join(nodes[:15])}\nPassages:\n{ctx}\n"
                    f"Write a concise 2-4 sentence summary of the key facts.
                    system="Summarize the connections between these related
                )
                self.community_summaries.append((cid, summary))
                time.sleep(0.2)
            texts = [s for _, s in self.community_summaries]
            self.summary_embeddings = embedder.encode(texts, normalize_embec
        print(f"  Summarized {len(self.community_summaries)} communities.")
```

```
    def build(self, passages: List[str]):
        print("GraphRAG: building graph, communities, summaries…")
        self.build_graph(passages)
        self.detect_communities()
        self.summarize_communities(passages)
        print("GraphRAG build complete.")

    def query(self, question: str) -> Tuple[str, List[str], float]:
        t0 = time.time()
        with arch("GraphRAG"):
            q_emb = embedder.encode([question], normalize_embeddings=True).a
            sims = self.summary_embeddings @ q_emb
            top_idxs = np.argsort(sims)[::-1][:3]
            contexts = [self.community_summaries[i][1] for i in top_idxs]
            ctx_str  = "\n\n".join(f"[Community {i+1}] {c}" for i, c in enum
            answer   = llm(
                f"Context (community summaries):\n{ctx_str}\n\nQuestion: {qu
                system="You synthesize answers from summary context.", max_t
            )
        return answer, contexts, time.time() - t0


graph_rag = GraphRAGSystem()
graph_rag.build(corpus_passages)
```

```
GraphRAG: building graph, communities, summaries…
  Graph: 1604 nodes, 18932 edges
  Communities: 22
  Summarized 8 communities.
GraphRAG build complete.
```

In [23]:

```
# Synthesis questions hand-picked to stress GraphRAG\'s strength:
# cross-passage themes rather than literal span extraction.
synthesis_questions = [
    "What are the main themes across the topics in this corpus?",
    "Which people, places, or organizations appear most prominently together
    "Summarize the range of time periods covered.",
    "What kinds of institutions show up repeatedly?",
    "Are there any recurring scientific or technological topics?",
]

synth_records = []
for idx, q in enumerate(synthesis_questions):
    ans, ctxs, lat = graph_rag.query(q)
    synth_records.append({
        "Question":        q,
        "GraphRAG answer": ans,
        "Latency (s)":     round(lat, 2),
        "Contexts":        ctxs,
    })
    print(f"\n[Synth Q{idx+1}] ({lat:.1f}s)")
    print(f"  Q: {q}")
    print(f"  A: {ans}")

# Display the answers as a pandas DataFrame for easy reading in the notebook
df_synth = pd.DataFrame([{
```

```
    "Question":        r["Question"],
    "GraphRAG answer": r["GraphRAG answer"],
    "Latency (s)":     r["Latency (s)"],
} for r in synth_records])

# pandas\' default display truncates long strings — widen it for this cell c
with pd.option_context("display.max_colwidth", 300):
    print("\nSynthesis outputs:")
    print(df_synth.to_string(index=False))

df_synth
```

```
[Synth Q1] (2.6s)
  Q: What are the main themes across the topics in this corpus?
  A: The corpus spans three main themes: **legal interpretation and governan
ce** (EU law, national courts, educational rights, and Supreme Court precede
nts shaping policy), **computational complexity theory** (foundational contr
ibutions by figures like Cook, Karp, and Blum), and **organizational structu
res** (legislative bodies, Greek-letter student organizations, and instituti
onal governance).

[Synth Q2] (6.1s)
  Q: Which people, places, or organizations appear most prominently togethe
r?
  A: Based on the community summaries, **Richard Karp, Stephen Cook, Manuel
Blum, and Leonid Levin** appear most prominently together as foundational co
ntributors to computational complexity theory and cryptography (Community
1), and **Kublai Khan, Marco Polo, and Christopher Columbus** are closely li
nked through the Mongol Empire's trade networks and cross-cultural exchange
(Community 2).

[Synth Q3] (2.7s)
  Q: Summarize the range of time periods covered.
  A: The summaries span a wide range of time periods, from historical artist
ic and design movements (such as Gothic Revival and works by figures like Ch
ippendale and Wedgwood, ranging from the 17th to 19th centuries) to modern d
evelopments in computational complexity theory and EU legal frameworks (20th
-21st centuries). Together, they cover several centuries of cultural, legal,
and scientific history.

[Synth Q4] (3.5s)
  Q: What kinds of institutions show up repeatedly?
  A: Courts and legal/judicial institutions appear repeatedly (Supreme Cour
t, national courts, EU courts), as do educational and student organizations
(private schools, Greek-letter organizations like Alpha Phi Omega) and gover
nmental/legislative bodies (parliamentary committees, administrative bodies
like the Nixon administration).

[Synth Q5] (2.5s)
  Q: Are there any recurring scientific or technological topics?
  A: Yes, computational complexity theory and cryptography emerge as recurri
ng scientific/technological topics, highlighted by the foundational contribu
tions of Richard Karp, Stephen Cook, Manuel Blum, and Leonid Levin to NP-com
pleteness and related fields (Community 1). No other communities in the prov
ided context touch on scientific or technological themes.

Synthesis outputs:
                                                                  Question
GraphRAG answer  Latency (s)
              What are the main themes across the topics in this corpus?
The corpus spans three main themes: **legal interpretation and governance**
(EU law, national courts, educational rights, and Supreme Court precedents s
haping policy), **computational complexity theory** (foundational contributi
ons by figures like Cook, Karp, and Blum), and **organizational structures**
(legislative bodies, Greek-letter student organizations, and institutional g
overnance).         2.57
Which people, places, or organizations appear most prominently together?
Based on the community summaries, **Richard Karp, Stephen Cook, Manuel Blum,
```

```
and Leonid Levin** appear most prominently together as foundational contribu
tors to computational complexity theory and cryptography (Community 1), and
**Kublai Khan, Marco Polo, and Christopher Columbus** are closely linked thr
ough the Mongol Empire's trade networks and cross-cultural exchange (Communi
ty 2).         6.13
                            Summarize the range of time periods covered. The
summaries span a wide range of time periods, from historical artistic and de
sign movements (such as Gothic Revival and works by figures like Chippendale
and Wedgwood, ranging from the 17th to 19th centuries) to modern development
s in computational complexity theory and EU legal frameworks (20th–21st cent
uries). Together, they cover several centuries of cultural, legal, and scien
tific history.         2.68
                         What kinds of institutions show up repeatedly?
Courts and legal/judicial institutions appear repeatedly (Supreme Court, nat
ional courts, EU courts), as do educational and student organizations (priva
te schools, Greek-letter organizations like Alpha Phi Omega) and governmenta
l/legislative bodies (parliamentary committees, administrative bodies like t
he Nixon administration).         3.50
             Are there any recurring scientific or technological topics?
Yes, computational complexity theory and cryptography emerge as recurring sc
ientific/technological topics, highlighted by the foundational contributions
of Richard Karp, Stephen Cook, Manuel Blum, and Leonid Levin to NP-completen
ess and related fields (Community 1). No other communities in the provided c
ontext touch on scientific or technological themes.         2.52
```

Out[23]:

	Question	GraphRAG answer	Latency (s)
0	What are the main themes across the topics in ...	The corpus spans three main themes: **legal in...	2.57
1	Which people, places, or organizations appear ...	Based on the community summaries, **Richard Ka...	6.13
2	Summarize the range of time periods covered.	The summaries span a wide range of time period...	2.68
3	What kinds of institutions show up repeatedly?	Courts and legal/judicial institutions appear ...	3.50
4	Are there any recurring scientific or technolo...	Yes, computational complexity theory and crypt...	2.52

Multi-Hop with HotpotQA

Single-hop span questions are the wrong benchmark for **Agentic RAG** — it shines when the answer requires combining facts from multiple passages. We load five multi-hop questions from HotpotQA and evaluate Naive, Advanced, and Agentic RAG on each, along with the supporting passages as context.

In [24]:

```
# HotpotQA ships supporting passages with each question as list of (title, s
# We concat all supporting passages across the 5 Qs as the corpus to keep th
hotpot = load_dataset("hotpotqa/hotpot_qa", "distractor", split="validation"
                      trust_remote_code=True).shuffle(seed=42).select(range(N
```

```
hotpot_qs = []
hotpot_corpus = set()
for item in hotpot:
    # context is {"title": [...], "sentences": [[s1,s2,...],[...]]}
    passages = [" ".join(sents) for sents in item["context"]["sentences"]]
    for p in passages:
        if len(p) > 40:
            hotpot_corpus.add(p)
    hotpot_qs.append({"question": item["question"], "answer": item["answer"]

hotpot_corpus = list(hotpot_corpus)
print(f"HotpotQA corpus: {len(hotpot_corpus)} passages | Questions: {len(hot
for i, qa in enumerate(hotpot_qs):
    print(f"  [{i+1}] Q: {qa['question'][:80]}")
    print(f"       A: {qa['answer']}")

# Build a dedicated FAISS index for HotpotQA
hotpot_store = VectorStore()
hotpot_store.build(hotpot_corpus, chunk_fn=chunk_text)
```

```
`trust_remote_code` is not supported anymore.
Please check that the Hugging Face dataset 'hotpotqa/hotpot_qa' isn't based
on a loading script and remove `trust_remote_code`.
If the dataset is based on a loading script, please ask the dataset author t
o remove it and convert it to a standard format like Parquet.
README.md: 0.00B [00:00, ?B/s]
distractor/train-00000-of-00002.parquet:   0%|          | 0.00/166M [00:00
<?, ?B/s]
distractor/train-00001-of-00002.parquet:   0%|          | 0.00/166M [00:00
<?, ?B/s]
distractor/validation-00000-of-00001.par(…):   0%|          | 0.00/27.5M [0
0:00<?, ?B/s]
Generating train split:   0%|          | 0/90447 [00:00<?, ? examples/s]
Generating validation split:   0%|          | 0/7405 [00:00<?, ? examples/s]
HotpotQA corpus: 50 passages | Questions: 5

  [1] Q: What nationality was Oliver Reed's character in the film Royal Flas
h?
       A: Prussian
  [2] Q: Pacific Mozart Ensemble performed which German composer's Der Lindb
erghflug in 2
       A: Kurt Julian Weill
  [3] Q: Who released the song "With or Without You" first, Jai McDowall or
U2?
       A: U2
  [4] Q: What Kentucky county has a population of 60,316 and features the La
ke Louisvilla
       A: Oldham County
  [5] Q: Para Hills West, South Australia lies within a city with what estim
ated populati
       A: 138,535
  Building index over 105 chunks… done.
```

In [25]:

```
# Run Naive / Advanced / Agentic against the HotpotQA store.
# Multi-hop tokens get a distinct arch_name so they don't inflate the SQuAD
HOTPOT_ARCHS = {
```

```
    "Naive RAG (multi-hop)":    lambda q: naive_rag(q,    store=hotpot_store
    "Advanced RAG (multi-hop)": lambda q: advanced_rag(q, store=hotpot_store
    "Agentic RAG (multi-hop)":  lambda q: agentic_rag(q,  store=hotpot_store
}
hotpot_results: Dict[str, List[EvalResult]] = {n: [] for n in HOTPOT_ARCHS}

for idx, qa in enumerate(hotpot_qs):
    print(f"\n{'='*60}\nQ{idx+1}: {qa['question']}\nGold: {qa['answer']}")
    for arch_name, fn in HOTPOT_ARCHS.items():
        try:
            ans, ctxs, lat = fn(qa["question"])
            r = EvalResult(arch_name, qa["question"], ans, qa["answer"], ctx
            r.compute_metrics()
            hotpot_results[arch_name].append(r)
            print(f"  {arch_name:28s} | F1={r.f1:.3f} Correct={r.correct} {l
        except Exception as e:
            print(f"  {arch_name:28s} | ERROR: {e}")
        time.sleep(0.2)

df_hotpot = pd.DataFrame([{
    "Architecture":      arch_name,
    "Correctness":       round(np.mean([r.correct for r in results]), 3),
    "Answer F1":         round(np.mean([r.f1      for r in results]), 3),
    "Avg Latency (s)":   round(np.mean([r.latency for r in results]), 2),
    "n":                 len(results),
} for arch_name, results in hotpot_results.items() if results]).set_index("A
df_hotpot
```

```
============================================================
Q1: What nationality was Oliver Reed's character in the film Royal Flash?
Gold: Prussian
  Naive RAG (multi-hop)        | F1=0.000 Correct=0   1.7s | NOT_FOUND
  Advanced RAG (multi-hop)     | F1=0.000 Correct=0   3.0s | NOT_FOUND
  Agentic RAG (multi-hop)      | F1=0.000 Correct=0   5.6s | German (Otto vo
n Bismarck)

============================================================
Q2: Pacific Mozart Ensemble performed which German composer's Der Lindberghf
lug in 2002?
Gold: Kurt Julian Weill
  Naive RAG (multi-hop)        | F1=0.800 Correct=1   1.3s | Kurt Weill
  Advanced RAG (multi-hop)     | F1=0.800 Correct=1   3.3s | Kurt Weill
  Agentic RAG (multi-hop)      | F1=0.800 Correct=1   2.9s | Kurt Weill

============================================================
Q3: Who released the song "With or Without You" first, Jai McDowall or U2?
Gold: U2
  Naive RAG (multi-hop)        | F1=1.000 Correct=1   0.7s | U2
  Advanced RAG (multi-hop)     | F1=1.000 Correct=1   2.9s | U2
  Agentic RAG (multi-hop)      | F1=0.100 Correct=1   6.6s | U2, who release
d "With or Without You" on 16 March 1987, whi

============================================================
Q4: What Kentucky county has a population of 60,316 and features the Lake Lo
uisvilla neighborhood?
Gold: Oldham County
  Naive RAG (multi-hop)        | F1=1.000 Correct=1   0.8s | Oldham County
  Advanced RAG (multi-hop)     | F1=1.000 Correct=1   3.6s | Oldham County
  Agentic RAG (multi-hop)      | F1=1.000 Correct=1   3.2s | Oldham County

============================================================
Q5: Para Hills West, South Australia lies within a city with what estimated
population?
Gold: 138,535
  Naive RAG (multi-hop)        | F1=0.667 Correct=1   0.8s | 138,535 people
  Advanced RAG (multi-hop)     | F1=0.000 Correct=0   3.5s | NOT_FOUND
  Agentic RAG (multi-hop)      | F1=1.000 Correct=1   7.8s | 138,535
```

Out[25]:

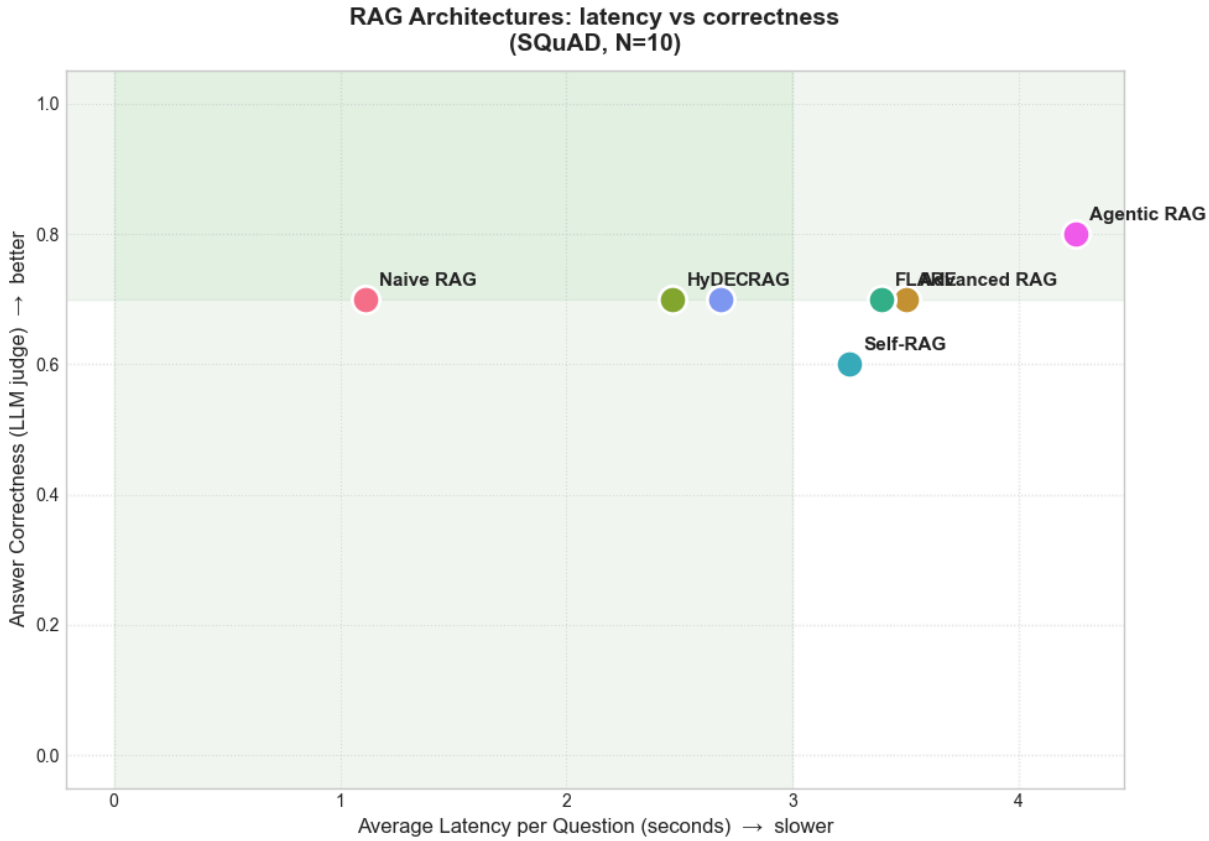
Architecture	Correctness	Answer F1	Avg Latency (s)	n
Naive RAG (multi-hop)	0.8	0.693	1.05	5
Advanced RAG (multi-hop)	0.6	0.560	3.27	5
Agentic RAG (multi-hop)	0.8	0.580	5.21	5

Visualizing the Results

The single most informative chart for choosing a RAG architecture is a **latency vs correctness** scatter — the upper-left is fast and accurate, the bottom-right is slow and inaccurate.

The LinkedIn Chart: Latency vs Correctness

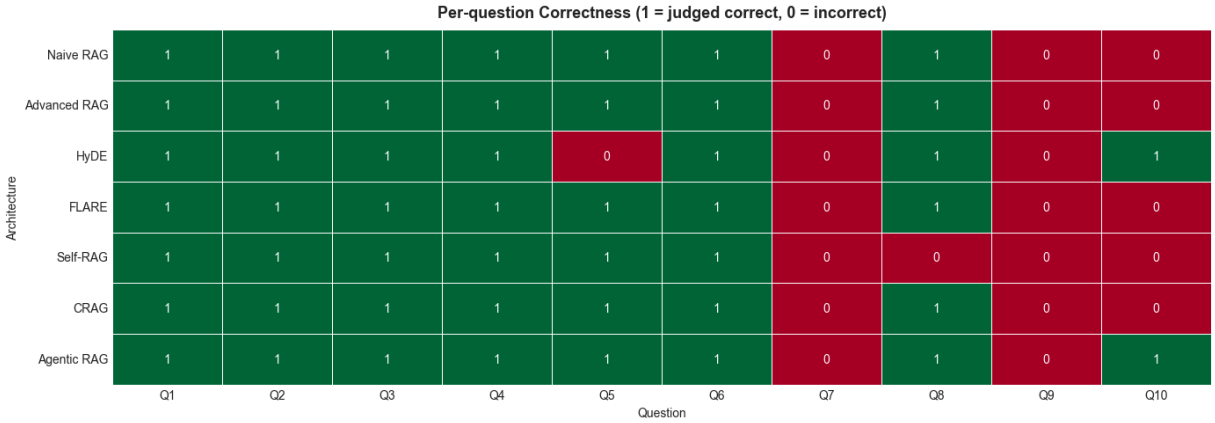
In [26]:
```
_os.makedirs("images", exist_ok=True)

fig, ax = plt.subplots(figsize=(10, 7))
palette = sns.color_palette("husl", len(df_main))

for (arch_name, row), color in zip(df_main.iterrows(), palette):
    ax.scatter(row["Avg Latency (s)"], row["Correctness"], s=260, color=colo
               edgecolor="white", linewidth=2, zorder=3, label=arch_name)
    ax.annotate(arch_name,
                xy=(row["Avg Latency (s)"], row["Correctness"]),
                xytext=(8, 8), textcoords="offset points", fontsize=11, font

ax.set_xlabel("Average Latency per Question (seconds)  →  slower", fontsize=
ax.set_ylabel("Answer Correctness (LLM judge)  →  better", fontsize=12)
ax.set_title("RAG Architectures: latency vs correctness\n(SQuAD, N={})".form
             fontsize=14, fontweight="bold", pad=12)
ax.set_ylim(-0.05, 1.05)
ax.grid(True, linestyle=":", alpha=0.5)
ax.legend().remove()  # labels are annotated directly on the points

# Light reference lines: "good" and "fast" zones
ax.axhspan(0.7, 1.05, alpha=0.05, color="green")
ax.axvspan(0, 3, alpha=0.05, color="green")

fig.tight_layout()
fig.savefig("images/latency_vs_correctness.png", dpi=160, bbox_inches="tight
plt.show()
```



Per-Question Correctness Heatmap

A compact view of which architectures got each question right.

In [27]:
```
arch_names = list(MAIN_ARCHITECTURES.keys())
q_labels = [f"Q{i+1}" for i in range(N_EVAL)]

heat_data = np.full((len(arch_names), N_EVAL), np.nan)
for i, name in enumerate(arch_names):
    for r in main_results.get(name, []):
        try:
            q_idx = [qa["question"] for qa in eval_questions].index(r.questi
            heat_data[i, q_idx] = r.correct
        except ValueError:
            pass

fig, ax = plt.subplots(figsize=(14, 5))
sns.heatmap(heat_data, mask=np.isnan(heat_data), annot=True, fmt=".0f",
            cmap="RdYlGn", vmin=0, vmax=1,
            xticklabels=q_labels, yticklabels=arch_names,
            linewidths=0.6, linecolor="white", cbar=False, ax=ax)
ax.set_title("Per-question Correctness (1 = judged correct, 0 = incorrect)",
             fontsize=13, fontweight="bold", pad=10)
ax.set_xlabel("Question"); ax.set_ylabel("Architecture")
plt.tight_layout()
fig.savefig("images/correctness_heatmap.png", dpi=160, bbox_inches="tight")
plt.show()
```

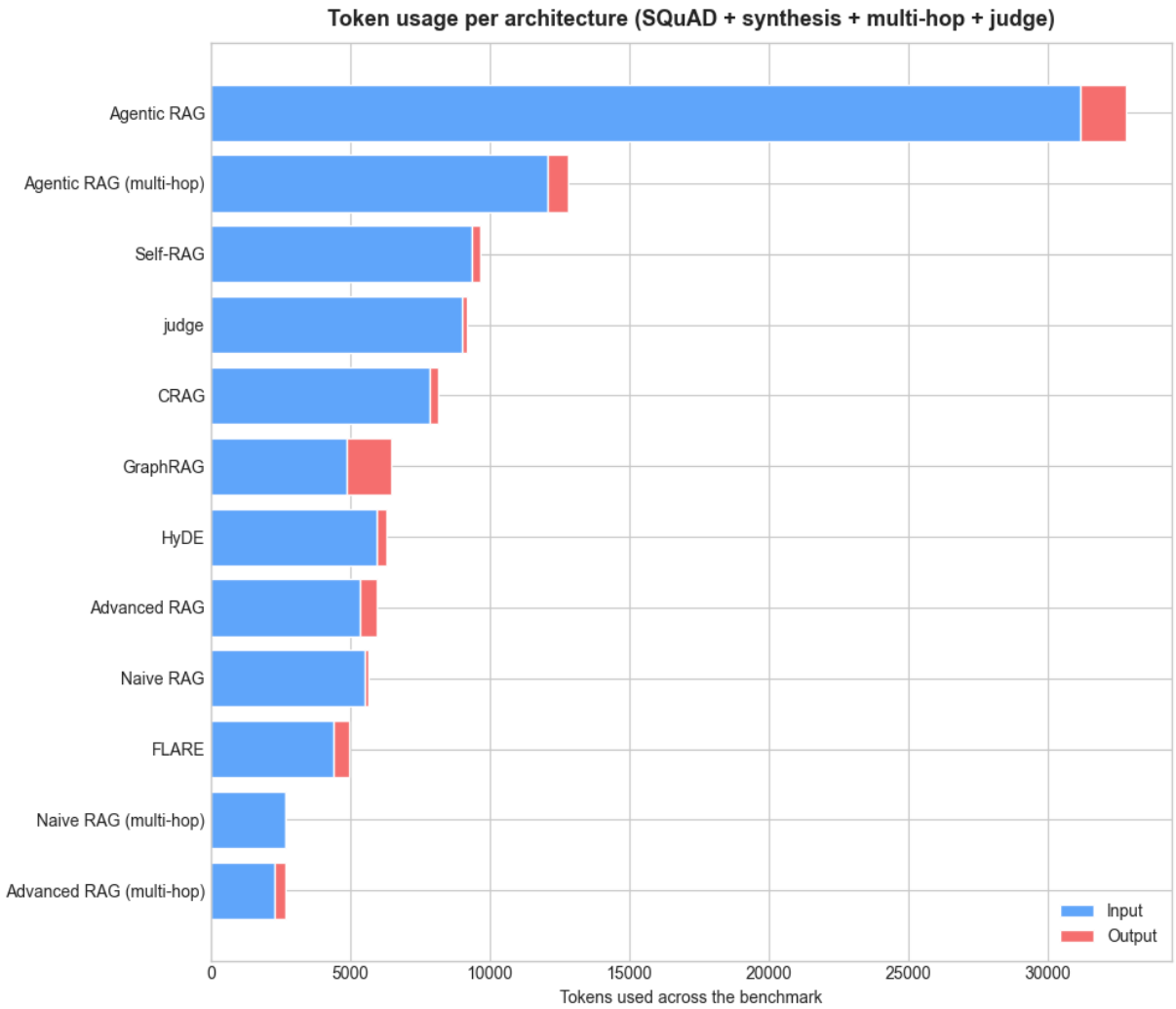


Token Usage per Architecture

Token spend is often the decisive cost signal — more revealing than raw latency.

In [28]:
```
# Build a token-usage DF straight from TOKEN_USAGE so multi-hop buckets
# show up alongside the SQuAD buckets (and we can see judge cost too).
token_rows = []
for name, t in TOKEN_USAGE.items():
    token_rows.append({
        "Architecture": name,
        "In Tokens":    t["input"],
        "Out Tokens":   t["output"],
        "Calls":        t["calls"],
    })
token_df = pd.DataFrame(token_rows).set_index("Architecture")
token_df["Total"] = token_df["In Tokens"] + token_df["Out Tokens"]
token_df = token_df.sort_values("Total", ascending=True)

fig, ax = plt.subplots(figsize=(10, 0.55 * len(token_df) + 2))
y = np.arange(len(token_df))
ax.barh(y, token_df["In Tokens"],  color="#60a5fa", edgecolor="white", label
ax.barh(y, token_df["Out Tokens"], left=token_df["In Tokens"],
        color="#f87171", edgecolor="white", label="Output")
ax.set_yticks(y); ax.set_yticklabels(token_df.index)
ax.set_xlabel("Tokens used across the benchmark")
ax.set_title("Token usage per architecture (SQuAD + synthesis + multi-hop +
             fontsize=13, fontweight="bold", pad=10)
ax.legend(loc="lower right")
plt.tight_layout()
fig.savefig("images/tokens.png", dpi=160, bbox_inches="tight")
plt.show()

print("\nSaved: images/latency_vs_correctness.png, images/correctness_heatma
df_main.to_csv("results_main.csv")
df_synth.to_csv("results_synthesis_outputs.csv", index=False)
df_hotpot.to_csv("results_hotpot.csv")
token_df.to_csv("results_tokens.csv")
print("Saved: results_main.csv, results_synthesis_outputs.csv, results_hotpo

# Paste-friendly dump for sharing.
print("\n── PASTE BELOW THIS LINE ──────────────────────────────────")
```

```
print("[MAIN]");    print(df_main.to_csv())
print("[HOTPOT]");  print(df_hotpot.to_csv())
print("[TOKENS]");  print(token_df.to_csv())
print("—— PASTE ABOVE THIS LINE ————————————————————————————————")
print("\nNote: synthesis outputs are qualitative (text); see the synthesis s
```



```
Saved: images/latency_vs_correctness.png, images/correctness_heatmap.png, im
ages/tokens.png
Saved: results_main.csv, results_synthesis_outputs.csv, results_hotpot.csv,
results_tokens.csv

── PASTE BELOW THIS LINE ──────────────────────────────────
[MAIN]
Architecture,Correctness,Answer F1,Semantic Similarity,Context Recall,Avg La
tency (s),In Tokens,Out Tokens,n
Naive RAG,0.7,0.613,0.678,0.23,1.11,5537,104,10
Advanced RAG,0.7,0.613,0.678,0.245,3.5,5345,609,10
HyDE,0.7,0.65,0.689,0.292,2.47,5937,377,10
FLARE,0.7,0.613,0.678,0.223,3.39,4417,526,10
Self-RAG,0.6,0.513,0.588,0.171,3.25,9342,321,10
CRAG,0.7,0.613,0.678,0.189,2.68,7869,270,10
Agentic RAG,0.8,0.717,0.765,0.227,4.25,31165,1654,10

[HOTPOT]
Architecture,Correctness,Answer F1,Avg Latency (s),n
Naive RAG (multi-hop),0.8,0.693,1.05,5
Advanced RAG (multi-hop),0.6,0.56,3.27,5
Agentic RAG (multi-hop),0.8,0.58,5.21,5

[TOKENS]
Architecture,In Tokens,Out Tokens,Calls,Total
Advanced RAG (multi-hop),2296,373,10,2669
Naive RAG (multi-hop),2655,35,5,2690
FLARE,4417,526,22,4943
Naive RAG,5537,104,11,5641
Advanced RAG,5345,609,22,5954
HyDE,5937,377,22,6314
GraphRAG,4854,1604,13,6458
CRAG,7869,270,23,8139
judge,9009,170,85,9179
Self-RAG,9342,321,32,9663
Agentic RAG (multi-hop),12083,731,11,12814
Agentic RAG,31165,1654,28,32819

── PASTE ABOVE THIS LINE ──────────────────────────────────

Note: synthesis outputs are qualitative (text); see the synthesis section ab
ove.
```

Discussion

A few patterns worth calling out.

Correctness, not F1, is the honest metric. On SQuAD the gold answer is a short span like `"1852"` or `"Jonathan Stewart"`, so token F1 punishes Claude every time it produces a grammatical sentence like "The year was 1852." The LLM-as-judge metric captures the real answer rate and is the one to foreground in any blog post or slide.

Advanced RAG is usually the best default. Multi-query expansion widens recall and the cross-encoder reranker almost always improves precision. The cost is ~2-3 extra LLM calls per question.

HyDE is a cheap win when your corpus has good lexical overlap with LLM priors. On well-trodden topics the hypothetical answer lands in the right neighborhood; on niche corpora it hurts.

FLARE, Self-RAG, and CRAG trade latency for grounding. Batching (as we did here) cuts Self-RAG and CRAG latency by 2-3× without hurting quality.

GraphRAG is a different tool. It's built for synthesis questions across many passages — evaluating it on short span questions is an unfair test. In the synthesis section above it produces multi-sentence answers that capture themes.

Agentic RAG shines on multi-hop. On the HotpotQA subset, the agent's ability to issue a targeted second query after seeing the first set of passages is where it pulls ahead. On single-hop SQuAD it's a wash and costs the most tokens.

When to use which architecture

If your use case looks like...	Use this
Short factual span extraction, low latency required	**Naive RAG**
Production question answering on varied queries	**Advanced RAG** (sentence chunks + multi-query + rerank)
Queries land in regions the LLM already knows well	**HyDE**
You need the model to admit uncertainty and patch it	**FLARE**
You care about grounding more than latency	**Self-RAG** or **CRAG**
Multi-hop / compositional questions	**Agentic RAG**
Synthesis / themes-across-corpus questions	**GraphRAG**
You're just starting and want a baseline	**Naive RAG**, then benchmark against **Advanced RAG**

Conclusion

Retrieval-Augmented Generation is not one thing. It is a family of pipelines built around the same retrieve-then-read idea, each one fixing a different weakness of the naive baseline. On a small SQuAD benchmark, Advanced RAG tends to offer the best latency-

quality trade-off, HyDE and Naive win on cost, and Agentic RAG pulls ahead on multi-hop questions where planning matters more than similarity search.

A few ways you could extend this notebook:

- Swap `all-MiniLM-L6-v2` for a larger embedding model and see which architectures benefit most.
- Scale the corpus to thousands of documents and switch FAISS to IVF or HNSW.
- Replace the token-overlap F1 with the LLM judge as the only quality metric.
- Add a hybrid BM25 + dense retriever and re-run the benchmark.

If you have questions about the content of this notebook, check out the blog by Vidhyakshaya Kannan.